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ORIGINAL RESEARCH

Early detection of students at risk of poor performance in Rwanda higher education using machine learning techniques

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Abstract The prediction of student performance is one of the major issues in many higher education institutions throughout the world. This issue is lately detected due to the lack of a proper continuous monitoring framework. Failing to detect and assist students leads them to the uncertainty of success. Therefore, there is a need to develop an early detection framework to identify students who are at risk of poor performance for a timely intervention. This paper aimed at proposing a Machine Learning based framework for the early detection of students at risk of poor performance in Higher Learning Institutions in Rwanda. To achieve this objective, selected machine learning models have been trained and tested using the dataset of both secondary school leavers data and higher institutions data. Since the data was imbalanced, the Synthetic Minority Oversampling Technique method has been used to get a balanced data set. Based on the accuracy evaluation metric criterion, the Decision Tree model has been chosen with 63.18% of accuracy and the model has been used to compute the feature importance diagram. Findings revealed that the High School Program is the best predictor of poor performance followed by the Senior 6 aggregate. The findings will assist educational policymakers and practitioners in regularly monitoring students who are at high risk of poor performance and in considering possible mentorship to help them perform better.

Keywords Performance · Machine learning · Prediction · Higher education

1 Introduction

The 2030 Agenda for Sustainable Development Goals lists quality education as a separate aim and places it at the center of its objectives [1]. Higher education is one of the most vital sectors that contribute to a country's economic growth and is fundamental to its long-term development. Several governments and non-governmental organizations in Sub-Saharan Africa have recently designed and launched policies to improve higher education quality and react positively to global challenges of qualified human capital[2].

The Government of Rwanda has been striving to achieve its predefined ambitious goals as well as millennium/sustainable development goals. Some of these goals such

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as Universal Primary Education, Gender Parity in Primary Education as well as empowering women by 2015 have been almost achieved [3]. In addition, Rwanda has established the National Strategy for Transformation 1. One of its priorities is to make the country a globally competitive knowledge-based economy [4]. To achieve this target, the country's educational quality must be enhanced. Rwanda aims at improving the quality of education by tackling dropout and class repetition among students as well as identifying the factors that impede students' progress from one level to another [3], [5].

Understanding the purpose of education and students' needs is critical for an educational system to be effective. Teachers, syllabus designers, curriculum designers, and the entire academic structure are all concerned about poor academic performance these days. Furthermore, absenteeism, carelessness, and infrequent college attendance are among the characteristics that are thought to be responsible for low academic performance in general. As an outcome, students are unable to understand what has been taught in class and are always behind the required performance level.

Poor performance in higher education is gradually becoming a global issue [6] and it is has become worse in poor countries especially in Sub-Saharan Africa. Consequently, producing skilled human resources for the market remains a challenge [7]. Therefore, one of the solutions to this challenge is to detect at an earlier stage the students at risk of poor performance [8], [9]. This detection helps in establishing intervention strategies that improve academic outcomes and prevent dropout. Those strategies may include the establishment of customized programs that provide adequate, effective interventions [10] and improved education standards [11], among others. For instance, once poor performance is early detected in a group of students, mentorship programs can be organized to support the concerned students promptly [12], [13].

Research has shown that Educational Data Mining is an essential approach to detect students at risk of poor academic outcomes [14–16] and the student's performance trends over periods can be monitored [11], thus helping in avoiding the dropout [17]. There is a considerable body of studies and research that aims to predict student performance as well as many other related academic outcomes of interest. In this regard, researchers have used different techniques to predict student performance at a given higher education institution. For example, Machine learning approaches have been used to predict students' performance based on their background and academic achievements over a term, and have proven to be a viable tool in predicting low and high performance at different education levels [15, 18].

By comparing various classification algorithms and pre-processing techniques, Akçapnar et al., [14] investigated which algorithms and features best predict the end-of-term academic performance of students and whether or not academic performance can be predicted using these features. According to the study's findings, 89% of failing students were correctly predicted by the kNN algorithm at the end of the term.

Koutina and Kermanidis [15] were curious to find out which machine learning method was most effective at forecasting the final grade of postgraduate informatics students at Ionian University. In comparison to other similar approaches, the results showed that the Naive Bayes and 1-NN models produced the best prediction outcomes.

To solve the problem of early prediction, Zhang and Rangwala [16], developed the Iterative Logistic Regression (ILR) technique. The suggested framework was able to efficiently forecast students who were likely to fail or drop out in upcoming semesters by utilizing past student records to the fullest extent.

González-Brambila et al., [18], provided an approach that made use of decision trees to forecast grades in two mathematics courses based on several performance metrics from a prior leveling course. Unfortunately, the resulting models lacked sufficient accuracy to be regarded as reliable for the intended use.

Hoffait and Schyns [19] targeted at early failure identification utilizing student data accessible at registration with a view to prompt and effective remediation and/or study reorientation. They contrasted three data mining techniques: random forest, logistic regression, and artificial neural networks. Their approach was able to detect a subset of 12.2% of students at risk of failure, with a 90% confidence rate.

Tsai et al., [14] employed machine learning techniques based on deep neural networks and statistical learning techniques to forecast the likelihood that students would drop out of school. The findings showed that a number of factors, including academic performance, applications for student loans, absence rates, and the number of subjects who were notified, could accurately predict whether or not a student would graduate from college.

Maksud and Nesar [20] modelled the student's knowledge level using ensemble methods to find out how early suggestions can be provided to the learners and the instructors. The findings showed that the combined voting ensemble (Fuzzy NN, NB, RF, and MLP) got 93.79% accuracy, which was higher than that of individual classifiers.

Aslam at al., [8] proposed a deep learning model to predict student performance to reduce the risk of failure. The data used contained socio-economic characteristics and course grades. Two courses (mathematics and Portuguese) served as reference. According to the results obtained, the

DL model achieved an accuracy of 0.964 and 0.932 for Portuguese Mathematics respectively.

In developed countries learning analytics has been integrated into education and it has proven successful in predicting learning outcomes [21], [14]. Many researches have focused on the application of machine learning approaches to early detection of students at risk of poor performance in many countries [19]. Nevertheless, none of these studies has looked at students' performance detection in Rwanda's Higher Learning Institutions.

The objective of this study is to propose a Machine Learning based framework for early detection of students at risk of poor performance in Higher Learning Institutions in Rwanda. Once detected, timely intervention can be administered to maintain a high retention rate and improve education outcomes [22].

This paper is organized as follows: after the introduction, sect 2 comprises the literature review, whereas sect 3 is composed of materials and methods followed by the results and discussion in sect 4, ended by the conclusion and references.

2 Literature review

The educational field has advanced in the recent decade, with the emergence of computer educational software, learning management systems, computerized assessment systems, and learning engagement [23]. Several factors influence students' poor academic performance. These include geographical factors, the historical and cultural background of students, educational approaches such as scary teaching techniques, inappropriate evaluation instruments, and students' basic knowledge. According to [24], biological, psychological, and social elements have an impact on a learner's performance.

Several studies have proposed machine learning-based approaches to detect students at risk of poor performance in higher education. Since complex input variables are expected, a key feature of machine learning is the ability to understand complex non-linear relationships. Examples of Machine Learning models considered in previous related studies include, but are not limited to, decision trees, Bayesian classifier-nearest neighbor, neural networks, support vector machines, random forest, logistic regression, linear discriminant analysis, multiple regression, and self-organized maps. The paragraphs below provide a comprehensive review of those researches.

Rowtho [25] built a system for identifying students who are at risk of poor performance at an undergraduate level in Mauritius. Regression analysis was used to identify variables that have a significant impact on academic success. The variables identified were personality, sex, year of study, learning methods, attendance rate, socio-economic status, and program of study. The author introduced a new technique that institutions might use to identify students who are at risk of poor performance and who could benefit from interventions prior to tests. This author made use of significant academic performance as evaluated by Grade Point Average (GPA) predictors.

Berens et al., [26] created an early detection system (EDS) that detects the students at risk of school dropouts in Germany using student administrative data and Machine Learning techniques. Predictors were social demographic data and academic performance data whereas the response was considered as graduate or dropout. Rather than relying on a single technique, the authors used neural networks, regression analysis, and the AdaBoost Algorithms to create an EDS. Based on students' academic scores and demographic variables, Ramanathan and Thangavel [27] applied three predictive models in which students were categorized as high, average, or low performers.

Using Machine Learning, Rashid and Aziz [28] attempted to find the relationship between students' course outcomes and their social backgrounds, previous achievements, and academic environments. Artificial neural networks were proposed in the authors' study to predict students' performance at the end of a General English course and they came up with a prediction accuracy of 80.9%.

Koutina and Kermanidis [15] investigated which machine-learning technique was the most successful in predicting students' final grades. Six well-known classification algorithms were evaluated. The best prediction results were attained using Naive Bayes and K-Nearest Neighbor (KNN) which were highly satisfactory when compared to similar approaches.

Four machine learning classifiers, namely Artificial Neural Networks (ANN), Naive Bayes, Decision Trees, and Logistic Regression were applied by Altabrawee et al., [29] to predict the performance of students in a computer science subject. This study focused on the impact of using the Internet as a learning resource as well as the impact of students' time spent on social media on their grades. The highest classification accuracy was 77.04% and it was achieved by ANN model.

Acharya and Sinha [12] examined the impact of internet use and social media time on students' performance. The ROC index performance evaluation metric and the classification accuracy were used to compare four machine learning models. The ANN model achieved the best performance compared to others.

Yağcı Mustafa [30] proposed a new model based on machine learning algorithms to predict the final exam grades of undergraduate students taking their midterm exam grades as the source data. The proposed model attained a classification accuracy ranging from 70–75%

according to the results. Babić [31] used artificial neural networks, classification trees, and support vector machines to predict students' performance with an accuracy of 65–100%. These results corroborate Ahmad and Shahzadi's findings [20] who evaluated students' study habits, learning skills, and academic interaction aspects and predicted students at risk for academic performance with 85% of accuracy.

Sandra et al., [32] analyzed 2753 articles from the IEEE Access and ScienceDirect databases between 2019 and 2021. They found out that nearly eight machine learning algorithms outperform the rest in predicting students' performance. These eight machine learning algorithms, namely Logistic Regression, Decision Trees, Random Forest, Adaboost, Support Vector Machine, K-Nearest Neighbor (KNN), Artificial Neural Network (ANN), and Naive Bayes Classifier (NBC) were selected for classification and tested using the data for this particular research.

Based on the existing research, various studies used demographic data and several others used student-related academic characteristics. The focus of this study is to employ school leavers' data and academic data in the context of Rwanda to predict the students at risk of poor performance in Higher Learning Institutions.

3 Materials and methods

3.1 Data

This study used data on secondary school leavers provided by Rwanda Education Board (REB) and data on students' academic scores from the University of Rwanda (UR) registrar's office. A designated team of enumerators extracted data from various files available in REB and UR. The sample includes 2783 observations on students enrolled in the University of Rwanda's Colleges of Science and Technology (CST) and College of Business and Economics (CBE). The sample was made of students who had sat for the national examination in 2014 and who had completed the first year in the two colleges in the academic year 2015–2016. The target variable was a binary variable built utilizing the number of failed modules in the first year of study as a measure of academic performance. When the number of failed modules is greater than one, it takes the value 1 (poor performance) and 0 (good performance) otherwise. The attributes used to predict the risk of poor performance include school location (rural or urban), senior six aggregate in the national examination, high school Program (STEM with Mathematics, STEM without Mathematics, and non-STEM), and age category (<19: Below normal age, [14, 16, 18–21]: Normal age, >25: Above normal age). Table 1 illustrates the sample distribution. There were 679 (24%) students with poor performance and 2104 (76%) students with good performance among the 2783 observations. This shows that the dataset is imbalanced.

Table 1 Sample distribution

Variable	Category	Number	%
S6 Aggregate	Continuous	2783	100
Age	< 19	72	3
	19–25	2600	93
	> 25	111	4
School location	Rural	1898	68
	Urban	885	32
High school program	Science without Math	230	8
	Non-STEM	801	29
	Science with Math	1752	63
Performance indicator	Poor performance (1)	679	24
	Good performance (0)	2104	76

3.2 Machine learning algorithms

Nowadays, organizations and businesses embrace machine learning to enhance decision-making through future prediction. Machine learning algorithms are generally classified into three categories: supervised learning, unsupervised learning, and reinforcement learning.

Mathematically, the research problem at hand is part of supervised learning where the motivation is to classify something or predict a value. Therefore, in this study, commonly used machine learning classification models [33] were trained using the data and their performances were compared using evaluation metrics such as accuracy, precision, recall, and F-score. The best eight machine learning algorithms highlighted in [32] were selected and used in this particular research as they emerged to perform well. The later are discussed below.

The logistic regression predicts the probability for classification problems with two possible outcomes. The positive class is predicted if the response variable meets or exceeds a discrimination threshold; otherwise, the negative class is predicted. The logistic function is used to model the response variable as a function of a linear combination of the explanatory variables.

The Decision trees are graphs that look like trees. A decision tree model's inference is computed by following the conditions while routing an example from the root (at the top) to one of the leaf nodes (at the bottom). The subsequent value of the leaf serves as the decision

tree's prediction. The set of visited nodes is known as the inference path.

A Random Forest [34] is a classifier that uses several decision trees on different subsets of the input dataset and averages the results to increase the dataset's prediction accuracy. While making predictions, random forests usually return the median or mean of the predictions made by the trees that make up the forest.

AdaBoost, short for Adaptive Boosting [33], is used as an ensemble method that combines multiple weak classifiers into a single strong classifier. As supervised learning, boosting is used to reduce bias and variance. The weights are reallocated to each instance, with higher weights being given to instances that were mistakenly classified, hence the name "adaptive boosting [34]."

A Support Vector Machine (SVM) [34] is a supervised learning algorithm that can be used for both regression and classification. It is more commonly used in classification problems. SVM solves complex classification, and outlier detection issues by performing all the best possible data transformations to establish boundaries between data points based on predetermined classes, labels, or outputs.

The K-nearest neighbors algorithm [33] also referred to as KNN or K-NN, is a supervised learning classifier that uses proximity to establish classifications or predictions about the grouping of a single data point. Although it can be applied to classification or regression problems, it is widely utilized as a classification algorithm. This is because it works according to the principle that similar points can be found close to one another [34].

Artificial neural network (ANN) [33] is an algorithm that copies the functions of the human brain and consists of several processing components that take inputs and generate results in accordance with predetermined activation functions. Neural networks are quicker than traditional computers in terms of processing data and they have better abilities for pattern detection and problem-solving [34].

The Naïve Bayes algorithm [34] is a supervised learning technique for classification problems that is based on the Bayes theorem. Being a probabilistic classifier, it makes a prediction based on the probability that an object will occur. It makes predictions about the probability that a given record or data point will belong to a specific class. The most likely class is the one with the highest probability.

3.2.1 Performance evaluation metrics

The Classification metrics evaluate a model's performance and indicate the reliability of the classification. These performance metrics let users evaluate how successfully the model performed for a given data [34].

Table 2 Sample of a confusion matrix

		Predicted	
		Positive	Negative
Observed	Positive	TP	FP
	Negative	FN	TN

Confusion Matrix—A Confusion matrix is a tabular representation of the ground-truth (observed) labels and model predictions (See Table 2). A confusion Matrix is not really a performance metric, but it serves as a sort of basis for how other metrics evaluate the outcomes.

Accuracy—The accuracy discusses the proportion of predictions that a classification model correctly predicted. It is given by Accuracy = (TP + TN)/(TP + FP + TN + FN), where

True Positive (TP) indicates how many positive class samples are correctly predicted by the model.

True Negative (TN) indicates how many negative class samples are correctly predicted by the model.

False Positive (FP) indicates how many negative class samples are incorrectly predicted by the model.

False Negative (FN) indicates how many negative class samples are incorrectly predicted by the model.

Precision—The precision is the proportion of correctly predicted negative or positive outcomes among all positively or negatively predicted cases.

Precision = TN/(TN + FN)

Recall—The recall is the proportion of positive predictions out of total actual positive outcomes.

Recall = TN/(TP + FN)

F1-score—The F1-score is a measure used to assess a binary classification model based on predictions for the positive class. As a result, the F1-score can be determined by taking the harmonic mean of both precision and recall and giving each variable equal weight.

F1-score = 2*[(precision*recall)/(precision + recall)]

ROC curve (Receiver Operating Characteristic curve)—The performance of a classifier model can be measured using the ROC curve metric. Since it compares two operating characteristics, the True Positive Rate (TPR) and the False Positive Rate (FPR), while the criterion changes, the ROC is also known as a relative operating characteristic curve. The ROC curve depicts the fraction of true positives to false positives, showing how sensitive the classifier model is. When the observed data is balanced throughout each class, ROC curve becomes more appropriate for use. TPR or recall is represented on the

Y-axis of the curve, whereas FPR is plotted on the X-axis, where

$TPR = TP/(TP + FN)$

$FPR = FP/(FP + TP)$

Area under curve (AUC) score—AUC is the area under the ROC curve. The AUC curve measure is generally used for binary classification problems. AUC calculates the two-dimensional area under the entire ROC curve from (0,0) to (1,1). The value of AUC ranges from 0 to 1. Therefore, a good model will have an AUC close to 1 and hence have a good degree of separability.

4 Results

The research experiments were done using the supervised machine learning algorithms that, according to the state of the art, perform well on learner analytics. The calculations and data modeling were done using Python software.

The dataset contains 679 (24%) students with poor performance against 2104 (76%) of good performance. This indicates that there was a class imbalance issue that needed to be addressed. To deal with such an issue, the synthetic minority over-sampling technique (SMOTE) was applied. The SMOTE approach was motivated by the fact that the dataset was small and dropping observations from the majority class would result in losing much information needed for model training.

The selected models were trained and tested using original and balanced data and data points were split into training and testing sets in the ratios of 80% and 20% respectively. This ratio is the commonly used one in machine learning studies. The same dataset conditions were applied to all selected machine learning algorithms. The fixed seed was applied to ensure that the randomly chosen train set and test sets are the same for all models. The advantage of this was to ensure appropriate models' comparison. The performance evaluation results of different models on test data are presented in Table 3. Results show that the performance metrics of all tested models on the class of interest were very small for the original dataset. For example, the ADABOOST and Decision Tree outperformed other models with a weighted average F1 score of 70%, but with F1 score of 22% and 24% respectively on the class of interest. This indicates that the imbalanced dataset leads to biased results towards the majority class. After dealing with data imbalance, the data points were 4208, and the three best performing models were the Decision Tree, the Random Forest, and the ADABOOST with accuracy percentages of 63.18%, 62.47% and 61.9%, respectively. The ROC–AUC for these three models is depicted in Fig. 2 and the AUC scores are presented in Table 4. The results in Table 4 demonstrate that the Adaboost has the lowest AUC score of 67% compared with the AUC score of 70% for Random Forest and AUC score of 70% for the Decision Tree. However, comparing the three models on the class of interest, the

Table 3 Performance evaluation results of different models on test data

Model		Precision		Recall		F1 score		Accuracy (%)	
		Imbalanced	Balanced	Imbalanced	Balanced	Imbalanced	Balanced	Imbalanced	Balanced
Adaboost	Weighted Avg	0.71	0.62	0.76	0.62	0.70	0.62	75.76	61.9
	Class of interest	0.51	0.63	0.14	0.59	0.22	0.61		
SVM	Weighted Avg	0.57	0.61	0.76	0.61	0.65	0.60	75.58	60.68
	Class of interest	0	0.64	0	0.48	0	0.55		
Random Forest	Weighted Avg	0.69	0.63	0.76	0.62	0.66	0.62	75.58	62.47
	Class of interest	0.43	0.63	0.02	0.59	0.04	0.61		
Naive Bayes	Weighted Avg	0.63	0.58	0.72	0.58	0.66	0.58	71.99	58.31
	Class of interest	0.24	0.60	0.07	0.52	0.10	0.56		
Logistic Regression	Weighted Avg	0.69	0.60	0.76	0.60	0.66	0.60	75.58	60.09
	Class of interest	0.50	0.61	0.01	0.55	0.03	0.58		
Decision Tree	Weighted Avg	0.71	0.64	0.76	0.59	0.70	0.62	75.58	63.18
	Class of interest	0.50	0.63	0.15	0.63	0.24	0.63		
Artificial Neural Network	Weighted Avg	0.57	0.59	0.76	0.59	0.69	0.59	75.58	0.59
	Class of interest	0	0.61	0	0.52	0	0.56		
KNN	Weighted Avg	0.66	0.58	0.66	0.57	0.66	0.57	69.53	57.48
	Class of interest	0.33	0.60	0.15	0.44	0.20	0.51		

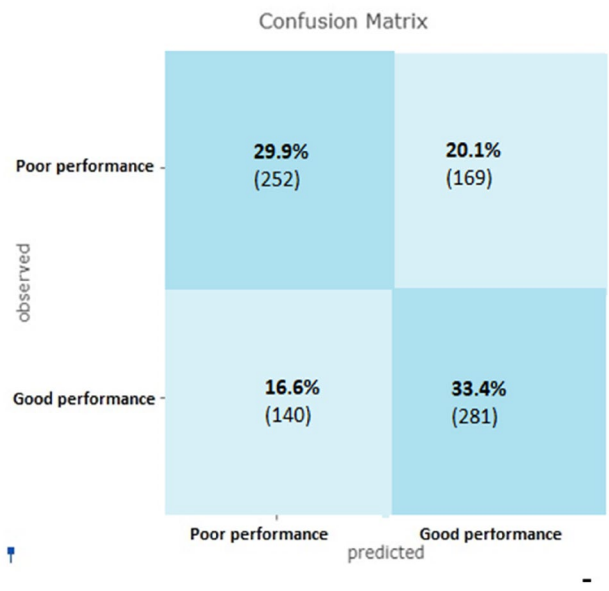

Fig. 1 Confusion matrix for decision tree

Decision Tree model emerged as the best performer on the basis of all considered performance evaluation metrics. The DT attained an Accuracy of 63.18% and a weighted average F1 score of 62%. On the class of interest, the F1-score increased from 24% on original data to 63% on balanced data. Therefore, the Decision Tree was selected as the best model in the current study.

The confusion matrix produced with the Decision Tree is depicted in Fig. 1. It shows that the Decision Tree can correctly predict a total of 533 = 252 + 281 (63.3%) cases.

The features of importance values plotted in Fig. 3indicate that the Decision Tree model ranked the features by their impact on the prediction of poor performance as follows: (1) high school program, (2) senior six aggregate, (3) school location and (4) age category.

5 Discussion

This section discusses the main results of the study. All the chosen algorithms have performed with an accuracy ranging between 57.48% and 63.18%. The top-performing model (Decision Tree) was selected based on its accuracy. By evaluating the performance of the three best models using Accuracy values, it was discovered that Decision Tree with 63.18% was the best-performing model. In previous studies, researchers also found similar findings on the capabilities of Decision Tree to predict performance in Higher Learning Institutions [12], [35]. Comprehensively and concisely, Ashraf et al., [35] carried out a comparative study of several recently used data mining techniques, classification algorithms, and their impact on datasets. Their findings showed that every data mining technique and algorithm produces different results depending on the dataset and

Table 4 Comparison between Decision Tree, Adaboost AUCs and Random Forest

Model	AUC Score
Decision tree	70%
Adaboost	67%
Random forest	70%

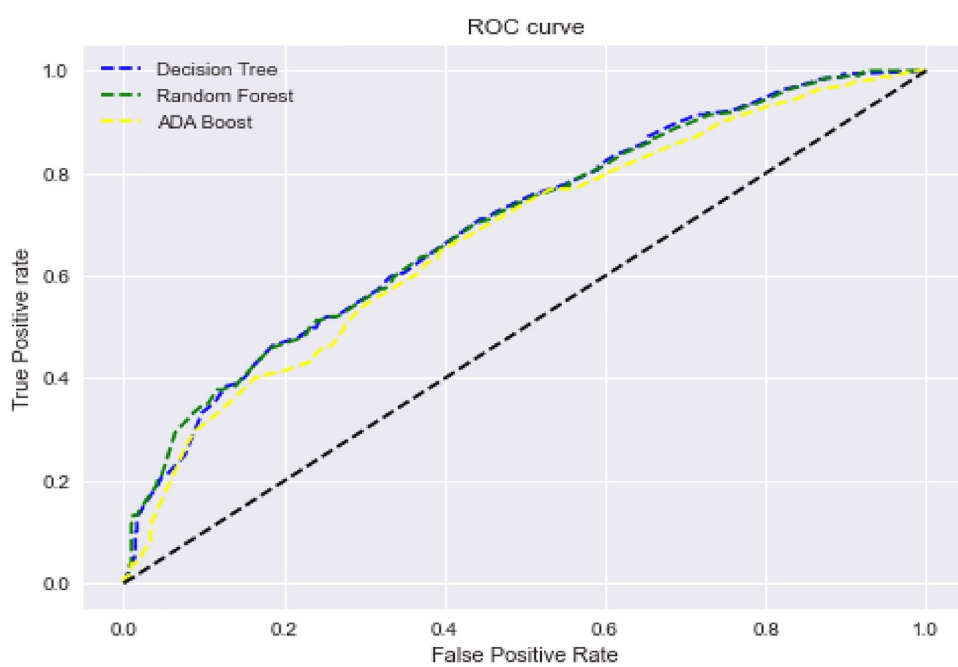

Fig. 2 ROC plot for the three chosen models

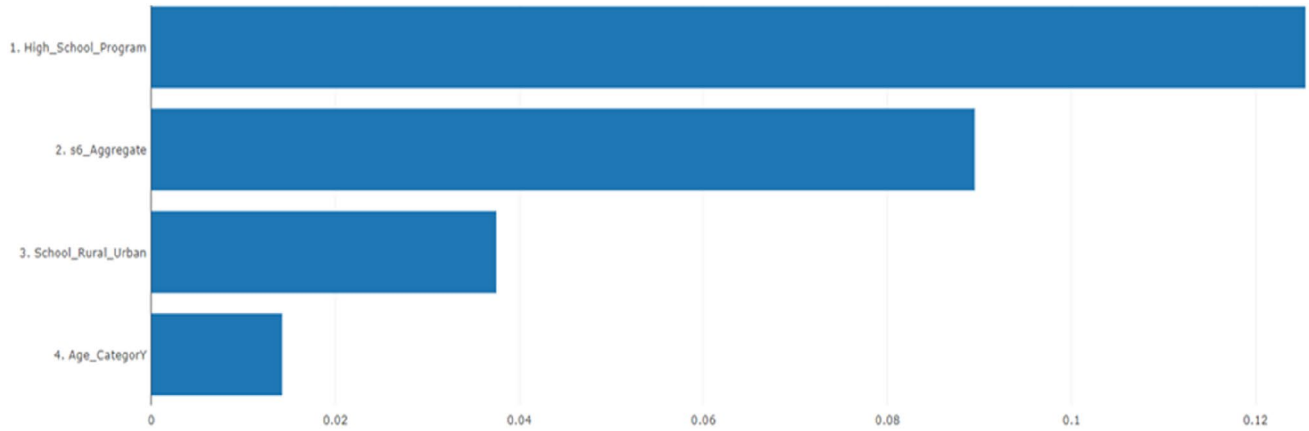

Fig. 3 Feature importances for decision tree

variable attributes being predicted. Using the cumulative grade point average (CGPA), demographic, background, and scholarship-related data, the decision tree algorithm was found to be the most effective.

Based on the results in previous studies, though it is hard to establish a concrete comparison due to different datasets in terms of size and features as well as study context, it is clear that the results of this study contribute more to the body of knowledge based on the types of analysis and experiments conducted to choose the best-unbiased model that could be deployed with capabilities of avoiding overfitting and underfitting. In addition to that, this is the first study that addresses issues related to the early detection of students at risk of poor performance in the context of Rwanda.

According to the findings, the following features were given priority in order of importance: high school program, senior six (S6) aggregate, school location, and age category. This implies that students' performance in higher education is significantly influenced by the program they followed in secondary school. This contradicts the findings of [36] where success before the university was determined as the potential key element that affects academic success at a higher learning level. These researchers used the ANN model and ranked academic success before entering the university (labeled 'S6 aggregate' in this paper) as the best predictor. In similar research, Akinleke [37] examined how socio-economic status affects the academic performance of college students. The results of the Chi-square and likelihood ratio tests demonstrated a clear correlation between high school and college performance and student satisfaction with the learning environment and support services. Similarly, adequate library services, extracurricular activities, and the distribution of college majors are consistently connected to college performance.

As far as ranking features by their impact on the prediction of poor performance is concerned, it is unexpected that the 'high school program' comes before the 'senior six aggregate'. It was revealed that programs with mathematics are very important determinants of performance in higher learning institutions in Rwanda. This result is consistent with the findings from a study by Islam and Al-Ghassani [38] which established that the student's performance in calculus in college is significantly predicted by their high school mathematics and general test scores. It is noted that this study selected the Decision Tree as the best model despite having an accuracy of 63.18 which may be considered small. This might be caused by the small and imbalanced dataset with fewer attributes.

6 Conclusion

This study aimed to predict earlier the students at risk of poor performance in Higher Learning Institutions in Rwanda using Machine Learning based framework. Secondary data from Rwanda Education Board (REB) on secondary school leavers and data from the University of Rwanda on students' academic scores have been used. The total number of observations was 2783. An adequate representation from eight Machine Learning models was trained and tested on the data to find the best model. The used dataset was split into two groups, the training set representing 80% and the testing set indicating 20%. The results proved that the three best models are the Decision Tree, the Random Forest, and the AdaBoost with accuracy of 63.18%, 62.47%, and 61.9% respectively. Based on the AUC score, both the Random forest and the Decision Tree emerged as the best model with an AUC score of 70% each. However, the Decision Tree was selected as it outperformed the Random Forest in predicting the

class of interest (poor performer). The selected model has been used for ranking features, and it turned out that the high school program is the most significant attribute to predict poor performance, followed by senior six (S6) aggregate, school location, and age category. This implies that students' performance in higher education is significantly influenced by the program they followed in secondary school and senior six aggregates obtained in the last national examinations. Moreover, instructors are thus in a position to spot the remarkable students as well as, more crucially, those who can score poorly in end-term examinations.

The results of this study will further highlight the significance of having a monitoring framework in higher education to identify students who perform poorly so as to arrange intervention measures for them. In addition, these findings are of paramount importance in assessing how enrollment to higher learning institutions should be conducted to reduce unnecessary dropouts and discontinuation and other failure-related effects resulting from poor performance. The findings from this study are in accord with those from previous studies. Therefore, educational decision-makers can apply the discovered model while establishing future policies.

The findings show that the high school program attribute has the highest impact on the performance of the students in higher learning studies and it is followed by the S6 aggregate. It should be imperative for the concerned authorities to consider all the given key performance influencing indicators while placing the students to higher learning institutions. This will help decision-makers to establish relevant policies that could enhance quality education for a highly-performing and successful academic society.

This study had some limitations mainly linked to the dataset with few available data points and few number of features. This may have impacted the overall performance of machine learning models.

This study focused on the University of Rwanda (UR), a public higher education institution. Further research will expand to the private higher education institutions in Rwanda, and more school leavers' characteristics will be taken into consideration.

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Data Availability The datasets used in this study can be obtained from the corresponding author upon request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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